CUSTOMER INSIGHT REPORT

Customer Shopping Behavior Analysis

Uncovering insights from customer purchases to guide strategic business decisions.



DATASET OVERVIEW

3,900 purchases reveal a broad behavior dataset

The dataset covers customer demographics and behavior, with a focused review of missing information.

3,900

total purchases

18

data columns covering customer demographics and behavior

37

missing values, all in the Review Rating column

DATA PREPARATION

The analysis moves from raw data to database-ready insight

A structured five-stage process turns the dataset into analysis-ready information.

Data loading

Import the dataset with pandas

Initial exploration

Check structure and summary statistics

Missing data handling

Impute Review Rating with median values

Feature engineering

Create age groups and purchase frequency

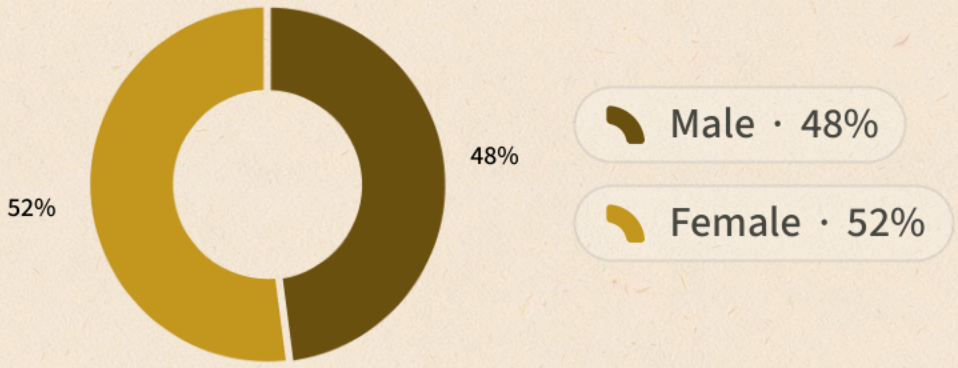
Database integration

Connect to **MySQL** for analysis

REVENUE MIX

Female customers generate slightly higher total revenue

Revenue is nearly balanced by gender, with female customers contributing a slightly larger share in the source analysis.



52%

Female revenue share

Male revenue share: 48%

DISCOUNT RESPONSE

Discounts attract customers who still spend above average

Smart shoppers

High spenders who maximize value with discounts

Strategic opportunity

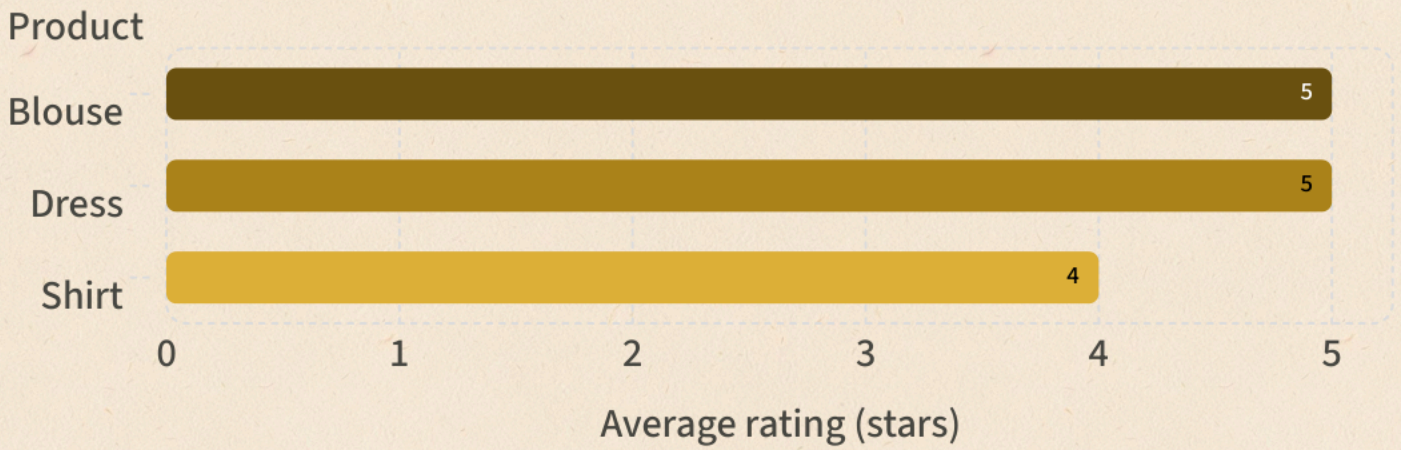
Target premium customers with exclusive offers



PRODUCT SATISFACTION

Blouses and dresses lead product satisfaction

Blouse and Dress earn the highest average rating in the source analysis.



5★

Blouse average rating

Dress average rating: 5 stars

Shirt average rating: 4 stars



SHIPPING BEHAVIOR

Express shipping customers spend more per transaction

Express shipping customers spend 12% more per transaction.



SUBSCRIPTION IMPACT

Subscribers combine higher spend with stronger loyalty

Subscription customers contribute a larger share of value and show a stronger repeat purchase pattern.

68%

higher spend among subscribers vs. non-subscribers

45%

of revenue comes from subscription customers

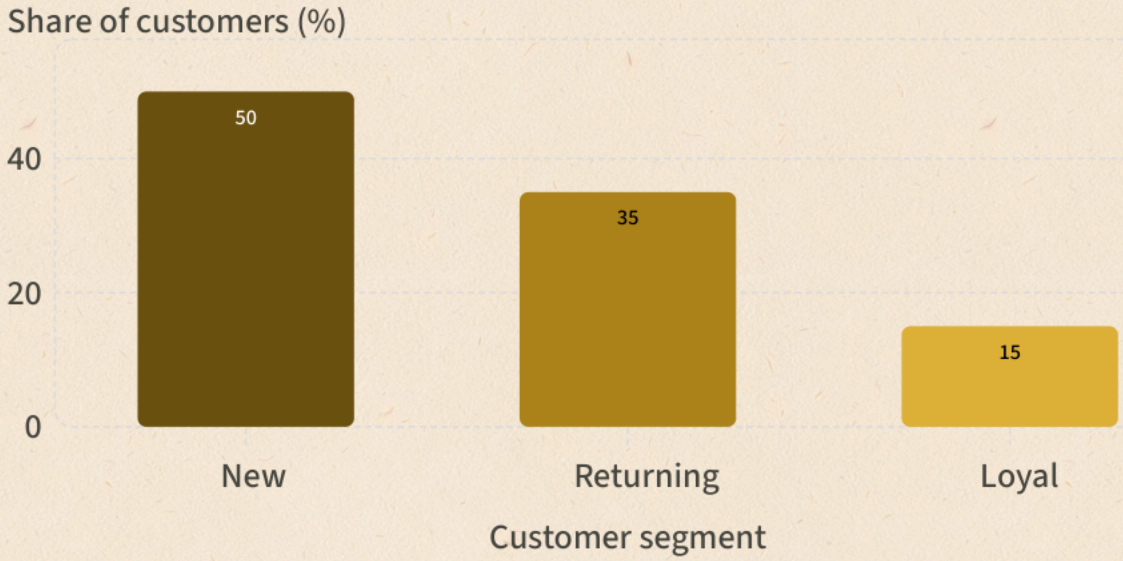
78%

loyalty rate based on repeat purchase frequency

CUSTOMER SEGMENTATION

Move customers from new to returning to loyal

Focus on converting New to Returning, and Returning to Loyal.







Thank you

Customer Shopping Behavior Analysis